

3.1 Waves + electromagnetic spectrum — 5054 Paper 2

Newest → oldest. Printed page numbers (especially 4) and graph ticks are not question numbers. One subquestion; MS is that part only.

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Question 5 (a)

(a) Fig. 5.1 represents the electromagnetic spectrum divided into the seven main regions.

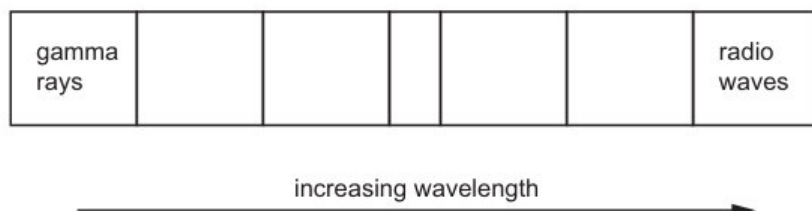


Fig. 5.1

The regions are arranged from left to right in order of increasing wavelength.

On Fig. 5.1:

- indicate the microwave region by marking it with the letter M
- indicate the ultraviolet region by marking it with the letters UV.

[1]

Mark scheme — Q5 (a) only

5(a)		B1
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5(a) B1

Question 5 (b)

(b) The frequency of the microwaves used by some satellite television systems is 12 GHz.

(i) The speed of microwaves in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of these microwaves in a vacuum.

wavelength = m [3]

(ii) Describe how microwaves are used in satellite television systems.

.....

 [2]

Mark scheme — Q5 (b) only

5(b)(i)	$(\lambda =) \frac{c}{f} \text{ or } \frac{3.0 \times 10^8}{1.2 \times 10^N}$	C1
	2.5×10^X (m)	C1
	0.025 (m)	A1
5(b)(ii)	encoded microwave / microwave signal / microwave carrying data / microwave transmitting data (to satellite from Earth)	B1
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	B1

5(b)(i) (■ =) c f or 8 3.0 10 1.2 10 N ■ ■ C1 2.5 ■ 10X (m) C1 0.025 (m) A1

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Question 5 (a)

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

- (ii) Describe the difference between a compression and a rarefaction.

.....

 [1]

Mark scheme — Q5 (a) only

5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1

5 Annotations guidance for centres Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component. We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an

Question 5 (b)

- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength [2]

Mark scheme — Q5 (b) only

5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	speed: no change	B1
	wavelength: decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1

5(b)(i) 0.75 (Hz) B1

Question 5 (c)

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(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]

Mark scheme — Q5 (c) only

5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1
------	---	----

5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 – Mark Scheme
PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 15 Question Answer Marks

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Question 4 (a)

- (a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

.....

..... [2]

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Mark scheme — Q4 (a) only

4(a)	motion of particles: backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions	B1

4(a) motion of particles: backwards and forwards / to and fro / closer and further apart B1 (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions B1

Question 4 (b)

- (b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

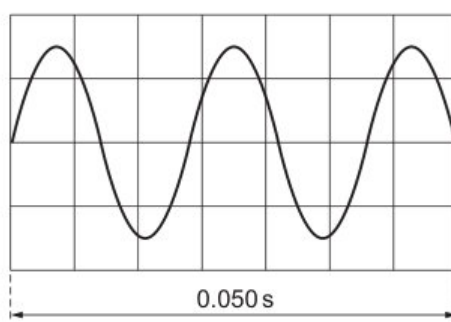


Fig. 4.2

The frequency of the sound is constant.

- (i) State what is meant by 'frequency'.

.....

..... [1]

- (ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.

DO NOT WRITE IN THIS MARGIN

Mark scheme — Q4 (b) only

4(b)(i)	number of oscillations per second or number of wave(length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f = 1 / T$ or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown and at least one complete oscillation seen	B1

4(b)(i) number of oscillations per second or number of wave(length)s (passing a point) per second B1

Question 4 (c)

- (c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

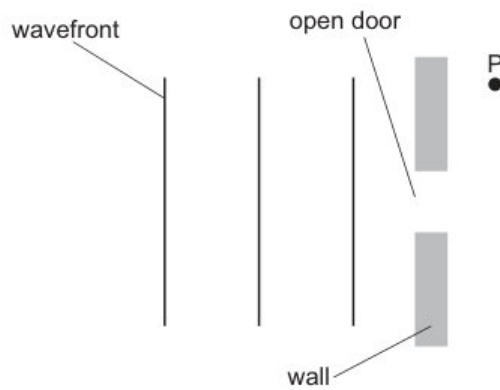


Fig. 4.4

- (i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]
- (ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]

Mark scheme — Q4 (c) only

4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left and some curving in correct direction	B1
4(c)(ii)	diffraction	B1

4(c)(i) (at least) three correctly curved wavefronts centered on gap B1 same wavelength as on left and some curving in correct direction B1

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Question 4 (a)

- (a) Draw on Fig. 4.1 to show the paths of the red light and the violet light between the left-hand side of the prism and the screen at point R and point V. [2]

Mark scheme — Q4 (a) only

4(a)	at least one ray from left-hand side showing correct refraction at both faces and no incorrect rays	B1
	at least two correct rays (i.e. two rays within glass) and one ending at R and the other at V	B1

4(a) at least one ray from left-hand side showing correct refraction at both faces and no incorrect rays B1 at least two correct rays (i.e. two rays within glass) and one ending at R and the other at V B1

Question 4 (b)

- (b) State what these observations show about the change in speed of red light and change in speed of violet light as each enters the glass from the air.

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN

Mark scheme — Q4 (b) only

4(b)	both (red and violet light) travel more slowly in glass than in air	B1
	red light travels more quickly (than violet light) in glass	B1

4(b) both (red and violet light) travel more slowly in glass than in air B1 red light travels more quickly (than violet light) in glass B1

Question 4 (c)

- (c) Five other colours of light appear on the screen between the red light at point R and the violet light at point V.

- (i) State the names of the five colours and list them in the correct order from red to violet.

red violet [2]

- (ii) State how the frequency and the wavelength of the coloured light change going from red light to violet light.

frequency

wavelength

[1]

DO NOT WRITE IN THIS MARGIN

Mark scheme — Q4 (c) only

4(c)(i)	any four (of the five) colours in order or five colours in any order	C1
	red orange yellow green blue indigo violet	B1

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4(c)(ii)	frequency increases and wavelength decreases	B1
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4 C1 28 (m / s) A1 2(b)(i) (■■■ =) ■■E / mc or 78.4 / (0.20 ■■ 1400) or gh / c or 9.8 ■■ 40 / 1400 C1 0.28 (°C) A1 2(b)(ii) some (thermal) energy is transferred to the ground or work is done deforming the modelling clay or air resistance / work done means that speed / k.e. is less than the calculated value B1 5054/21 – Mark Scheme PUBLISHED October/November 2024 © Cambridge University Press & Assessment 2024 Page 6 of 10 Question Answer Marks 3(a) vibrating atoms / particles (ignore electrons) collide with electrons B1 electrons move (a long distance) through metal B1 (moving) electrons collide / interact with distant atoms (ignore electrons) or electrons carry energy (through metal) B1 3(b)(i) their / the particles' (average) speed increases B1 3(b)(ii) the particles collide (with each other / the walls of the container) B1 they / the parti

Question 4 (d)

(d) On Fig. 4.1, point P is shown immediately next to point R.

No light is observed at point P, but a detector at point P registers radiation reaching the detector.

State which region of the electromagnetic spectrum is detected at point P.

..... [1]

[Total: 8]

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Mark scheme — Q4 (d) only

4(d)	infrared (radiation)	B1
------	----------------------	----

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1

4(d) infrared (radiation) B1 Question Answer Marks

Question 5 (b)

(b) Describe how a longitudinal wave differs from a transverse wave.

.....
.....
.....
..... [2]

Mark scheme — Q5 (b) only

5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1

5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1

Question 5 (c)

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency

wavelength

[2]

[Total: 9]

Mark scheme — Q5 (c) only

5(c)(i)	($\lambda = v / f$ or $1500 / 8.4 \times 10^6$)	C1
	1.8×10^{-4}	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

5 C1 0.49 (N) A1 1(b)(ii) ($W = Fx$ or 0.49×0.86 C1 0.42 (J) A1 1(b)(iii) the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases) B1 the work done in one second / unit time increases B1 Question Answer Marks 2(a)(i) (G.P.E. =) mgh or 0.20×9.8 C1 78 (J) A1 2(a)(ii) (K.E. =) $\frac{1}{2}mv^2$ or $\frac{1}{2} \times 0.2 \times v^2$ C1 ($v = \sqrt{2gh}$ or $2 \times 9.8 \times 40$ or 784 C1 28 (m / s) A1 2(b)(i) ($E = mc^2$ or $78.4 / (0.20 \times 1400)$ or gh / c or $9.8 \times 40 / 1400$ C1 0.28 (°C) A1 2(b)(ii) some (thermal) energy is transferred to the ground or work is done deforming the modelling clay or air resistance / work done means that speed / k.e. is less than the calculated value B1 5054/21 – Mark Scheme PUBLISHED October/November 2024 © Cambridge University Press & Assessment 2024 Page 6 of 10 Question Answer Marks 3(a) vibrating at

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Question 4 (a)(i)

- (i) State what is meant by 'frequency'.

.....
..... [1]

- (ii) Calculate the speed of ultrasound in air.

Show your working.

speed = m/s [2]

- (iii) State **one** use of ultrasound.

..... [1]

Mark scheme — Q4 (a)(i) only

4(a)(i) number of (complete) waves (passing a point) per second B1

Question 4 (a)(ii)

- (ii) Calculate the speed of ultrasound in air.

Show your working.

speed = m/s [2]

- (iii) State **one** use of ultrasound.

..... [1]

Mark scheme — Q4 (a)(ii) only

4(a)(i) number of (complete) waves (passing a point) per second B1

Question 4 (b)

(b) Sound waves are longitudinal waves. Water waves are transverse waves.

- (i)** Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

.....

.....

.....

.....

..... [3]

Mark scheme — Q4 (b) only

4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1

Question 4 (c)

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.

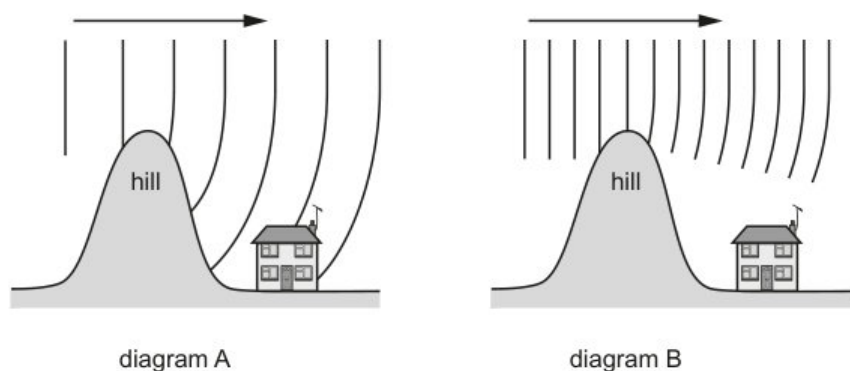


Fig. 4.1 (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

.....

.....

.....

..... [2]

Mark scheme — Q4 (c) only

4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

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Question 4 (a)

(a) Calculate the wavelength in air of this red light.

wavelength = m [3]

Mark scheme — Q4 (a) only

4(a)	$(\lambda =) v / f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1

4(a) (■ =) v / f or 3.0 ■ 108 / 4.7 ■ 1014 C1 6.4 ■ 10N (m) or 3 / 4.7 ■ 106 C1 6.4 ■ 10−7 (m) A1

Question 4 (b)(i)

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed

frequency

wavelength

[2]

- (ii) Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = [3]

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Mark scheme — Q4 (b)(i) only

4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1

4(b)(i) speed: decreases frequency: stays the same wavelength: decreases B2

Question 5 (a)

- (a) The name of one of these three regions is the ultraviolet region.

- (i) State the names of all three regions in order of increasing frequency.

smallest frequency

intermediate frequency

largest frequency

[2]

- (ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect

.....

property

.....

[2]

Mark scheme — Q5 (a) only

5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

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5(a)(i) infrared (radiation) and visible (light) and ultraviolet (radiation) B1 infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order B1

Question 5 (b)

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

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Mark scheme — Q5 (b) only

5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	$(R =) V / I$ or 1.5 / 0.20 or 7.5 (Ω)	C1
	$(R_T =) 7.5 - 1.5 - 3.5$	C1

5(b) in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction B1 in a longitudinal wave, the vibration direction is parallel to the propagation direction B1 Question Answer Marks

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Question 6 (a)

(a) State what is meant by 'frequency'.

.....

..... [2]

Mark scheme — Q6 (a) only

6(a)	(frequency =) the <u>number</u> of wavelengths	B1
	(passing a point) per <u>unit</u> time	B1

6(a) (frequency =) the number of wavelengths B1 (passing a point) per unit time B1

Question 10 (b)

(b) The gamma radiation produced has a frequency of 8.8×10^{19} Hz.

The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = m [2]

[Total: 9]

Mark scheme — Q10 (b) only

10(b)	$(\lambda =) v / f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	3.4×10^{-12} (m)	A1

10(b) (■ =) $v / f = 3.0 \div 8.8 \times 10^{19}$ C1 3.4×10^{-12} (m) A1

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Question 7 (a)

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

Mark scheme — Q7 (a) only

7 (A) B1 3(b) ($P = I^2 R$ or VI and either $V = IR$ or $V = IR$ numerical or algebraic in any form C1 or voltage across 400 Ω resistor = (a) 400 (6.857 V) 0.12 (W) A1 3(c) circuit drawn with all 3 resistors in parallel with cell / power supply B1 ammeter in series with 100 Ω resistor or in series with resistor with largest current in arrangement drawn B1 5054/21 – Mark Scheme PUBLISHED May/June 2023 Page 8 of 13 Question Answer Marks
 4(a)(i) ($P =$) Force / Area algebraic or numerical C1 520 000 (Pa) A1 4(a)(ii) They are equal (in size and direction) B1 4(a)(iii) small(er) area of contact (with board) C1 large(r) pressure (on board) A1 4(b)(i) pressure (of water) act / pressure in bag B1 force (on water) at right angles to the surface / in a straight line or pressure acts in all directions B1 4(b)(ii) larger pressure at A / greater weight of w

Question 7 (b)

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.



Fig. 7.2

[2]

Mark scheme — Q7 (b) only

7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1

Question 7 (c)

- (c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm.

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s.

- (i) Define the term 'wavelength'.

.....
..... [1]

- (ii) Calculate the frequency of the wave.

frequency = Hz [2]

- (iii) Calculate the wavelength of the wave in the shallow part of the tank.

Mark scheme — Q7 (c) only

7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1

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Question 5 (a)

- (a) Underline **two** other examples of transverse waves.

seismic P-waves seismic S-waves sound X-rays [1]

Mark scheme — Q5 (a) only

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5(a) seismic S-waves and X-rays B1

Question 5 (b)

- (b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

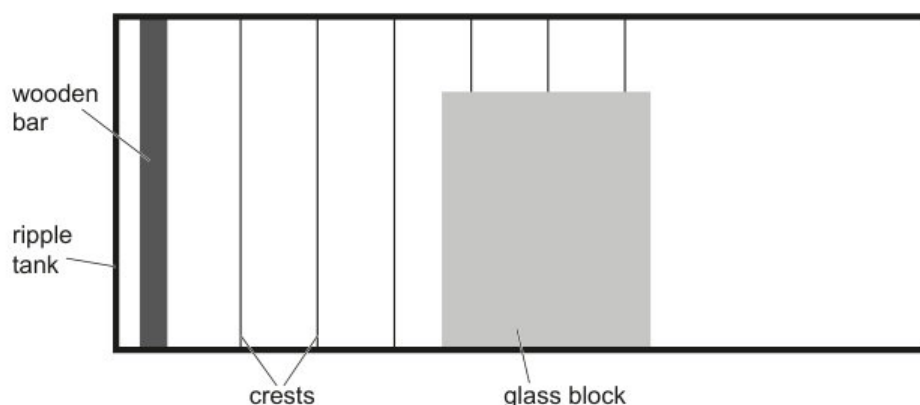


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

- (i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

Mark scheme — Q5 (b) only

5(b)(i) (wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$ C1 frequency 6.7 (Hz) A1 wavelength 4.0 (cm) A1

Question 7 (a)

- 7 (a) Ultraviolet radiation is one component of the electromagnetic spectrum.

- (i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1

2 [1]

- (ii) State **one** useful application of ultraviolet radiation.

..... [1]

- (iii) Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

..... [1]

Mark scheme — Q7 (a) only

7(a)(i) X-rays and gamma (radiation / rays) B1

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Question 4 (a)

- (a) State the name of the spectrum of waves that includes X-rays.

..... [1]

Mark scheme — Q4 (a) only

4(a)	electromagnetic (waves) or components of electromagnetic spectrum	B1
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4(a) electromagnetic (waves) or components of electromagnetic spectrum B1

Question 4 (b)

- (b) (i) Explain what is meant by 'frequency'.

.....

 [2]

- (ii) State the name of the waves in this spectrum that have the greatest frequency.

..... [1]

Mark scheme — Q4 (b) only

4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1

4(b)(i) number of oscillations / cycles / wavelengths B1 per unit time B1

Question 4 (c)

- (c) X-rays are used in hospitals to produce an image of broken bones.

- (i) Explain how this is done. You may draw a diagram if you wish.

.....

 [3]

- (ii) Explain why X-rays are **not** used for pre-natal scanning.

.....
 [1]

[Total: 8]

Mark scheme — Q4 (c) only

4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage or X-rays show the bone structure rather than foetal development or image would not be clear	B1

4(c)(i) X-rays directed towards body part with (suspected) broken bone B1 X-rays pass through flesh but not (to same extent) through bones B1
image produced by digital / electronic / CCD / CMOS detector B1

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Question 4 (b)

(b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....

 [2]

Mark scheme — Q4 (b) only

4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1

4(b) (they) vibrate M1 (vibration) parallel to / in the direction of propagation / of energy travel A1

Question 4 (c)

(c) Describe what is meant by 'a rarefaction'.

.....
 [1]

Mark scheme — Q4 (c) only

4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1
------	--	-----------

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Question 8 (a)

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

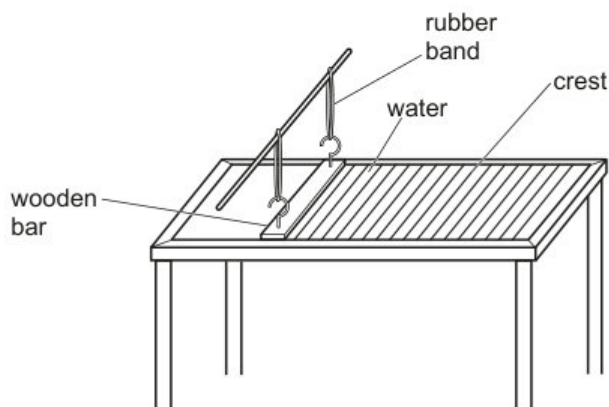


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

- (i) Calculate the number of crests produced in 1.5 s.

number = [1]

- (ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

Mark scheme — Q8 (a) only

8(a)(i) 3 B1

Question 8 (b)

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
 [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed
 wavelength [2]

Mark scheme — Q8 (b) only

8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1

Question 8 (c)

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

..... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

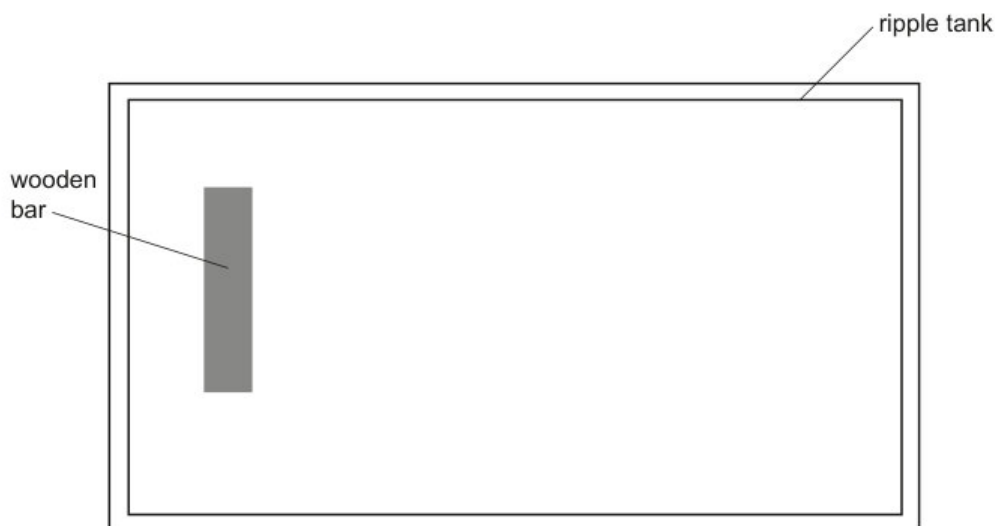


Fig. 8.3

Mark scheme — Q8 (c) only

8 cm3 B1 5054/22 – Mark Scheme PUBLISHED May/June 2022 Page 7 of 12 Question Answer Marks 4(a) clear angle is to / from the normal or 90°– angle between ray and surface C1 angle between/with the refracted ray / beam and the normal or vv or angle from the normal to the refracted ray / beam or vv A1 4(b) (refractive index =) $\sin i / \sin r$ algebraic or numerical in any form M1 1.5 A1 4(c) refractive index smaller (for red) B1 larger angle of refraction (for red) or (red) bends / refracts less (towards normal) assume comparison is for light entering glass at equal incident angles unless clear otherwise B1 4(d) P infrared and Q ultraviolet B1 Question Answer Marks 5(a) middle box of 1st three boxes and top box of 2nd three boxes B1 5(b)(i) (P =) VI or I 2R algebraic or numerical seen in any form C1 0.0024 W A1 5(b)(ii) V = IR algebraic or nu

2021

Oct/Nov 5054/21

Question 8 (b)(iii)

- (iii) Sound is a longitudinal wave.

Describe the difference between a longitudinal wave and a transverse wave.

.....

.....

.....

..... [2]

- (iv) Using Fig. 8.2, determine the number of times that the cone reverses its direction of motion in 1.0 s.

number = [2]

- (v) The speed of sound in air is 340 m/s.

Calculate the wavelength of the sound.

Mark scheme — Q8 (b)(iii) only

8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 Page 9 of 11
Question Answer Marks

Question 8 (b)(v)

- (v) The speed of sound in air is 340 m/s.

Calculate the wavelength of the sound.

wavelength = [2]

Mark scheme — Q8 (b)(v) only

8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 Page 9 of 11
Question Answer Marks

Question 3 (a)

(a) Light travels more slowly in the block than in air.

(i) Explain how Fig. 3.1 shows this.

.....

 [2]

(ii) State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength
 frequency [2]

Mark scheme — Q3 (a) only

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection). 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted. 5 'List rule' guidance For questions that require n responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked ignore in the mark scheme should no

Question 4 (a)

4 (a) Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.



Fig. 4.1

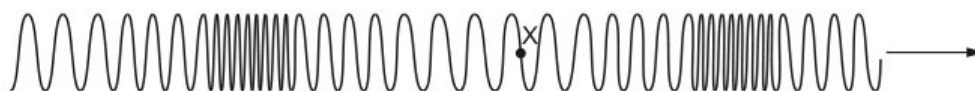


Fig. 4.2

(i) State the name of each of the two types of wave shown.

Fig. 4.1
 Fig. 4.2 [1]

(ii) On Fig. 4.1, draw a wave which has a smaller frequency than the wave shown. [1]

(iii) Describe the movement of point X on the spring in Fig. 4.2 as the wave passes.

.....
 [1]

Mark scheme — Q4 (a) only

May/June 5054/22

Question 10 (a)

- 10 (a)** Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

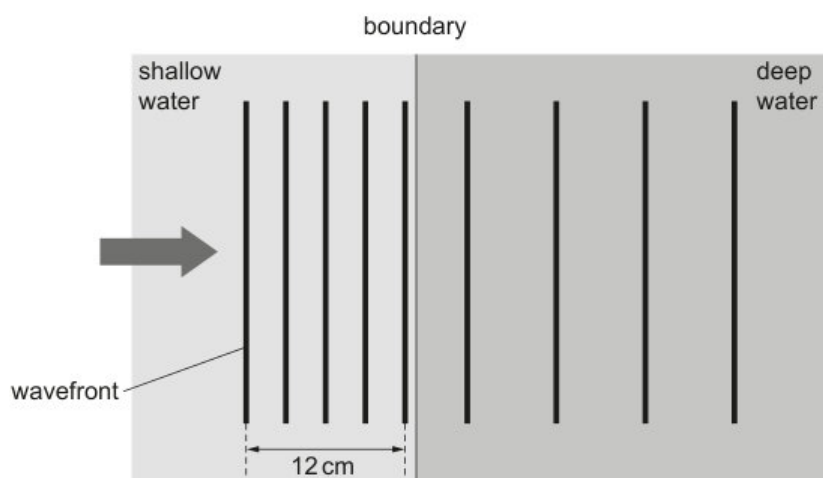


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (i) Determine the wavelength of the wave in the shallow water.

wavelength = [1]

Mark scheme — Q10 (a) only

10(a)(i) 3 (.0) cm B1

Question 10 (b)

(b) White light is made up of different colours.

- (i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength

.....

.....

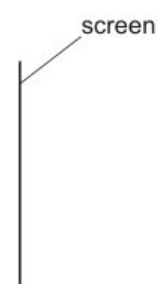
largest wavelength

[2]

- (ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.



Mark scheme — Q10 (b) only

10(b)(i) four colours from violet, indigo, blue, green, yellow, orange, red B1 correct order of four of the above colours B1

2020

Oct/Nov 5054/21

Question 4 (a)

- (a) Light travels more slowly in glass than in air.

- (i) State what happens to the wavelength of the light in the ray as it enters the glass.

..... [1]

- (ii) State what happens to the frequency of the light in the ray as it enters the glass.

..... [1]

Mark scheme — Q4 (a) only

4(a)(i) it decreases B1

May/June 5054/21

Question 9 (a)(ii)

- (ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....
 [2]

- (iii) To produce the image of an unborn child, an ultrasound emitter and receiver are placed close together on the mother's skin.

Fig. 9.1 shows pulses detected by the receiver.

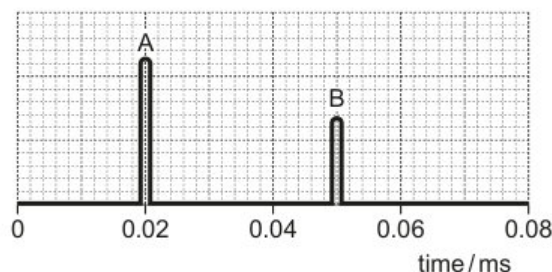


Fig. 9.1

Pulse A is the emitted pulse and pulse B is the first pulse that returns from the unborn child.

The average speed of ultrasound in human tissue is 1500 m/s.

Calculate the distance between the emitter and the child.

Mark scheme — Q9 (a)(ii) only

9(a)(i) (sound that) cannot be heard C1 (sound with) frequency above 20 kHz A1

May/June 5054/22

Question 5 (a)

- (a) Draw **one** line from each component of the spectrum to another suitable use for that component.

component	use
	sunbeds
microwaves	television controller
	satellite television
ultraviolet	airport security check of cases

[2]

Mark scheme — Q5 (a) only

5(a) microwaves →satellite television B1 ultraviolet →sunbeds B1

Question 10 (b)

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.

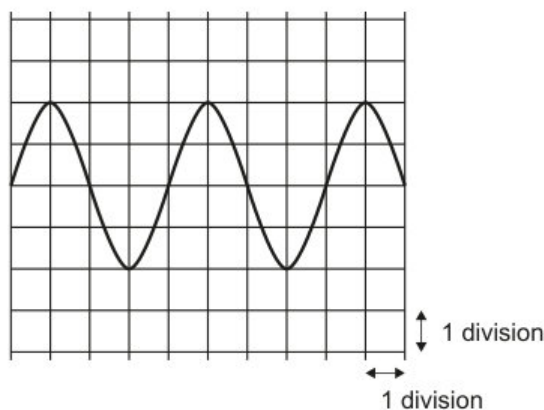


Fig. 10.1

The settings on the oscilloscope are 10 ms/division for the x-axis and 3.0 V/division for the y-axis.

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5054/22/M/J/20

Mark scheme — Q10 (b) only

10(b)(i) two divisions C1 6(.0) V A1

Question 10 (c)(i)

- (i) State what must happen to the metal filament for thermionic emission to occur.

.....
 [1]

- (ii) Explain why the electrons accelerate away from the filament.

.....

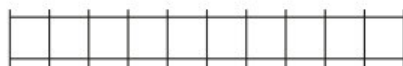
 [1]

- (iii) The charge on one electron is $1.6 \times 10^{-19} \text{ C}$.

Calculate the maximum kinetic energy of one electron after it has been accelerated through 2000 V.

kinetic energy = [2]

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.



Mark scheme — Q10 (c)(i) only

10(c)(i) longitudinal B1 5054/22 – Mark Scheme PUBLISHED May/June 2020 Page 13 of 13 Question Answer Marks

Question 10 (c)(ii)

- (ii) Explain why the electrons accelerate away from the filament.

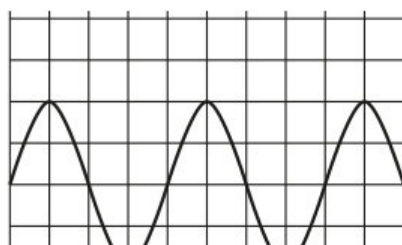
.....
.....
..... [1]

- (iii) The charge on one electron is $1.6 \times 10^{-19} \text{ C}$.

Calculate the maximum kinetic energy of one electron after it has been accelerated through 2000 V.

kinetic energy = [2]

- (b) Fig. 10.1 is the trace on the screen of the oscilloscope.



Mark scheme — Q10 (c)(ii) only

10(c)(i) longitudinal B1 5054/22 – Mark Scheme PUBLISHED May/June 2020 Page 13 of 13 Question Answer Marks

2019

Oct/Nov 5054/21

Question 4 (a)

- (a) State the name of the component labelled:

- A
- B
- C [2]

Mark scheme — Q4 (a) only

4(a) (A:) ultraviolet (radiation) (B:) infra-red (radiation) (C:) microwaves any two correct C1 all three correct A1

Question 4 (b)

- (b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency
the smallest wavelength [1]

Mark scheme — Q4 (b) only

Question 10 (a)

- (a) State the speed of light in a vacuum.

..... [1]

Mark scheme — Q10 (a) only10(a) $3.0 \times 10^8 \text{ m/s}$ B1**Question 10 (b)**

- (b) State the colours of the visible spectrum.

..... [1]

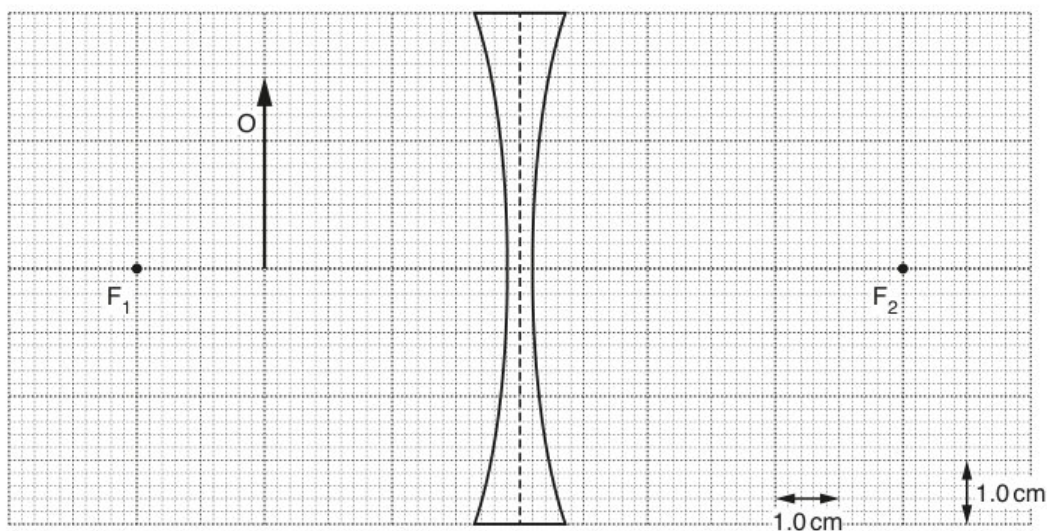
Mark scheme — Q10 (b) only

10(b) red orange yellow green blue indigo violet (any order) B1

Question 10 (c)

- (c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focusses), F_1 and F_2 , of the lens.

**Fig. 10.1**

- (i) Determine the focal length of this lens.

focal length = [1]

Mark scheme — Q10 (c) only10(c)(i) $(-)$ 6.0 cm B1 5054/21 – Mark Scheme PUBLISHED October/November 2019 Page 10 of 10 Question Answer Marks**Question 10 (d)**

- (d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

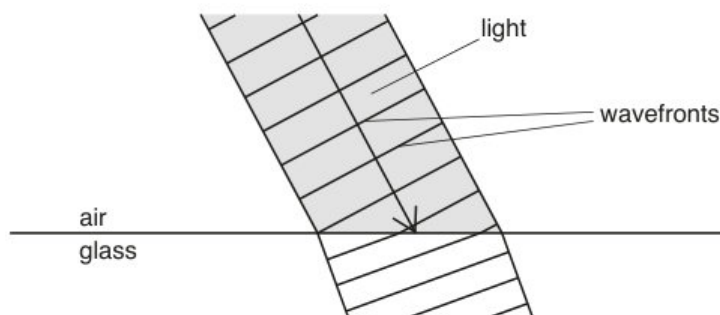


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

Mark scheme — Q10 (d) only

10(d) light travels more slowly in glass or light changes speed B1 one side / left-hand side of wavefront slows down first B1 wavelength decreases or wavefront travels a shorter distance in the same time B1

Oct/Nov 5054/22

Question 5 (a)

- (a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

Mark scheme — Q5 (a) only

5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 – Mark Scheme PUBLISHED October/November 2019 Page 6 of 10 Question Answer Marks

Question 5 (b)(i)

- (i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

- (ii) Describe how the ultrasound cleans the jewellery.

.....
.....
..... [2]

[Total: 6]

Mark scheme — Q5 (b)(i) only

5(b)(i) ($\lambda =$) v / f or 1500 / 42 000 C1 0.036 m or 3.6 cm A1

May/June 5054/21

Question 10 (a)

- (a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

- (ii) Define the term *frequency*.

.....
..... [2]

- (iii) Determine the frequency of the wave.

frequency = [1]

Mark scheme — Q10 (a) only

10 A range for 220 and 320 mA readings and 0–200 mA range for 100 mA reading or start with 10 A / highest range and either • reduce to 200 mA for last reading • reduce range and keeping high value on scale / not over scale B1 5054/21 – Mark Scheme PUBLISHED May/June 2019 Page 7 of 11
Question Answer Marks 6(a) blows / melts / cuts off circuit / stops current and when the current is high B1 6(b)(i) $I = P/V$ numerical or algebraic B1 6.2, 6.25 or 6.3 A B1 6(b)(ii) any integral value 7–13 A B1 6(c) double insulated or outside case / body is plastic / rubber B1
Question Answer Marks 7(a)(i) magnetic field / flux mentioned B1 change in field / flux (in coil) or field (lines) cut coil / wire B1 7(a)(ii) more turns or stronger magnet or move magnet faster B1 7(b)(i) correct symbol B1 7(b)(ii) converts more of the input energy / power into lig

May/June 5054/22

Question 4 (a)

- (a) Determine the wavelength where the radiation is brightest.

wavelength = [1]

Mark scheme — Q4 (a) only

4(a) $1.15\text{--}1.25 \times 10^{-6}$ (m) B1

Question 4 (b)

- (b) Visible light has a wavelength between 4.0×10^{-7} m and 7.0×10^{-7} m.

The radiation in (a) lies just outside the visible part of the electromagnetic spectrum.

State the name of the region of the spectrum that contains the radiation in (a).

..... [1]

Mark scheme — Q4 (b) only

4(b) infra-red B1

Question 4 (c)

- (c) Determine the frequency of the radiation in (a).

The speed of light is $3.0 \times 10^8 \text{ m/s}$.

frequency = [3]

[Total: 5]

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5054/22/M/J/19

[Turn over

Mark scheme — Q4 (c) only

4(c) ($f = v / \lambda$) algebraic or numerical C1 $3 \times 10^8 / 1.2 \times 10^{-6}$ C1 $2.5 \times 10^{14} \text{ Hz}$ A1 5054/22 – Mark Scheme PUBLISHED May/June 2019 Page 6 of 10 Question Answer Marks

2018

Oct/Nov 5054/21

Question 9 (a)

- (a) State how a longitudinal wave differs from a transverse wave.

.....
..... [1]

Mark scheme — Q9 (a) only

9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1

Question 9 (b)

- (b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

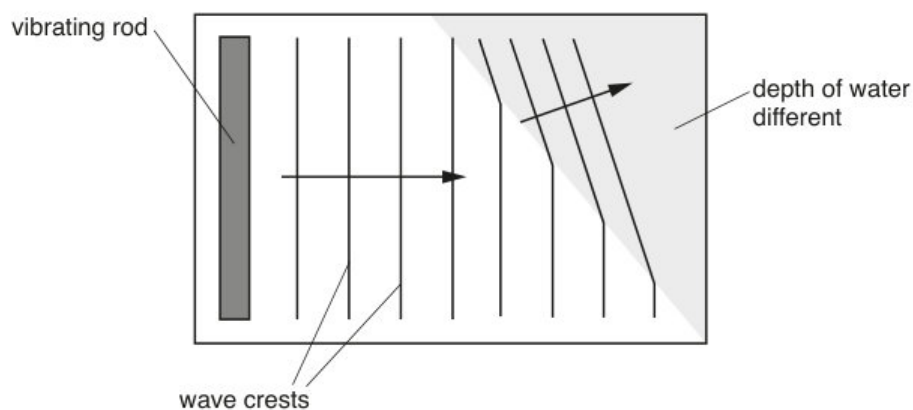


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

- (i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

Mark scheme — Q9 (b) only

9(b)(i) ($f = v / \lambda$ or $0.17 / 0.019$ C1 8.9 Hz A1

Oct/Nov 5054/22

Question 9 (b)

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

- (i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....

.....

.....

.....

..... [3]

- (ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.

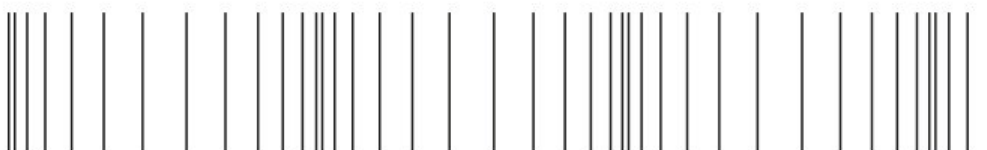


Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]

Mark scheme — Q9 (b) only

9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1

Question 9 (c)

(c) A sound wave travelling in a liquid, passes into air.

- (i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

Mark scheme — Q9 (c) only

9 C1 1200 m / s A1

May/June 5054/21

Question 9 (a)

- (a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

	true	false
Gamma rays are used to kill cancerous cells but can also cause cancer.	☐	☐
Infra-red is used in sun-beds.	☐	☐
Radio waves have the highest frequency in the electromagnetic spectrum.	☐	☐
The higher the frequency of the radiation the smaller is the wavelength in air.	☐	☐

[2]

Mark scheme — Q9 (a) only

9(a) true false false true all 4 correct 2 marks B1 3 correct 1 mark B1

Question 9 (c)

- (c) The Earth is 1.5×10^8 km from the Sun.

- (i) State the speed of light in a vacuum.

.....[1]

- (ii) Calculate the time taken for light to travel from the Sun to the Earth.

time =[2]

Mark scheme — Q9 (c) only

9(c)(i) 3.0×10^8 m / s B1

Question 11 (a)

- (a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) The water wave in the ripple tank transfers energy.

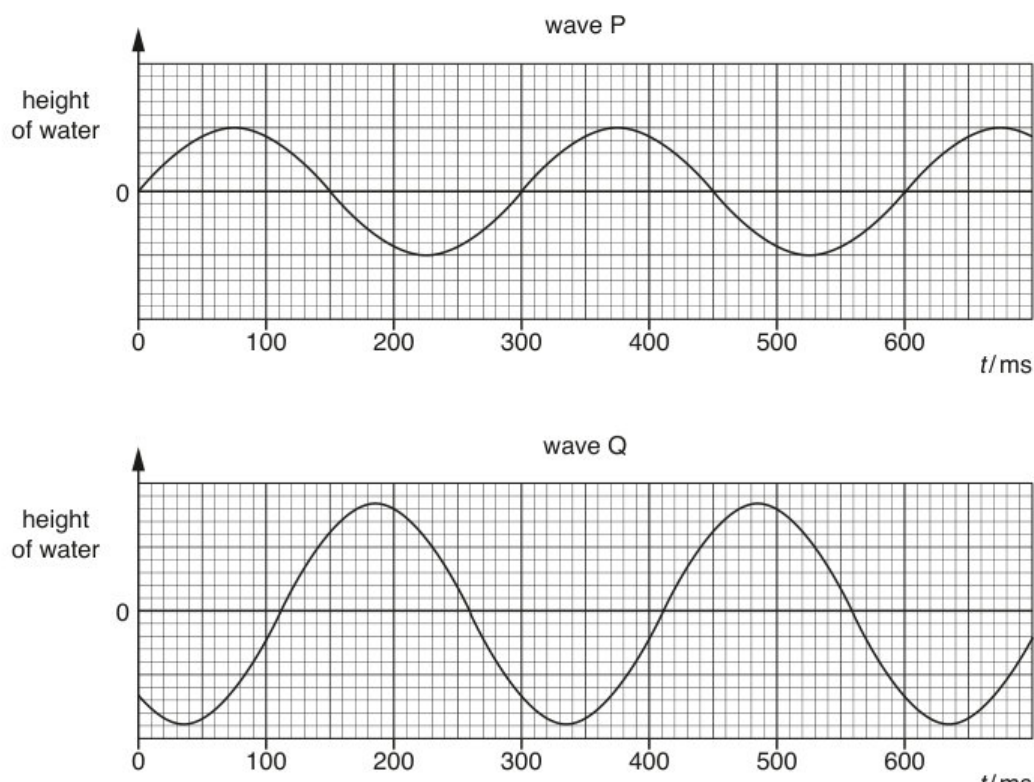
.....

Mark scheme — Q11 (a) only

11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1

Question 11 (b)

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.



Mark scheme — Q11 (b) only

11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1

May/June 5054/22

Question 5 (b)

- (b) The microwave signal is used in a telephone system that produces sound.

Waves are described as longitudinal or transverse. Some waves are also electromagnetic.

- (i) Place ticks (✓) in the table in Fig. 5.1 to show whether microwaves and sound waves are longitudinal or transverse and to show whether they are electromagnetic.

	longitudinal	transverse	electromagnetic
microwaves			
sound			

[2]

Fig. 5.1

- (ii) A longitudinal wave contains a compression.

Describe what is meant by a *compression*.

.....

[1]

Mark scheme — Q5 (b) only

5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1

2017

Oct/Nov 5054/21

Question 10 (a)

- (a) State the name of the component of the electromagnetic spectrum that is found in region

- (i) P,
 (ii) Q,
 (iii) R, [2]

Mark scheme — Q10 (a) only

10(a) P - gamma(-rays) or γ (-rays) Q - ultraviolet (radiation) R - microwaves any one correct C1 all three correct A1

Question 10 (b)

- (b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

Mark scheme — Q10 (b) only

10(b) P and X-rays and Q ticked B1

Question 10 (c)

- (c) The speed of electromagnetic radiation in a vacuum is $3.0 \times 10^8 \text{ m/s}$.

A television remote controller uses infra-red radiation of wavelength $9.4 \times 10^{-7} \text{ m}$.

- (i) Calculate the frequency of this radiation.

frequency = [3]

- (ii) Explain how infra-red radiation is used in television remote controllers.

.....
.....
.....
..... [2]

Mark scheme — Q10 (c) only

10(c)(i) ($f = v/\lambda$ or $3.0 \times 10^8 / 9.4 \times 10^{-7}$ C1 3.2×10^8 C1 $3.2 \times 10^{14} \text{ Hz}$ A1

Oct/Nov 5054/22

Question 6 (a)

- (a) State what is meant by the term *echo*.

.....
..... [1]

Mark scheme — Q6 (a) only

6(a) reflection of sound B1

Question 6 (b)

- (b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (i) the amplitude of the sound wave,

..... [1]

- (ii) the frequency of the sound wave.

..... [1]

Mark scheme — Q6 (b) only

6(b)(i) decreases B1

Question 6 (c)

- (c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength =[2]

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5054/22/O/N/17

Mark scheme — Q6 (c) only

6(c) ($\lambda =$) c / f or $330 / 3700$ C1 0.089 m or 8.9 cm or 89 mm A1 5054/22 – Mark Scheme PUBLISHED October/November 2017 Page 6 of 10
Question Answer Marks

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Question 10 (b)

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

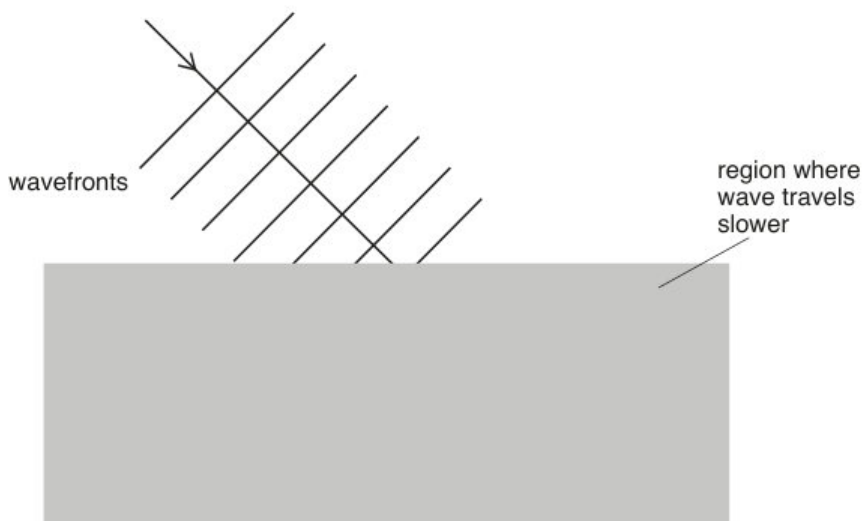


Fig. 10.2

- (i) State what is meant by *wavefront*.

.....

101

Mark scheme — Q10 (b) only

10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1

Question 10 (c)(i)

(i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

(ii) Apart from its position, state one characteristic of the image.

.....[1]

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

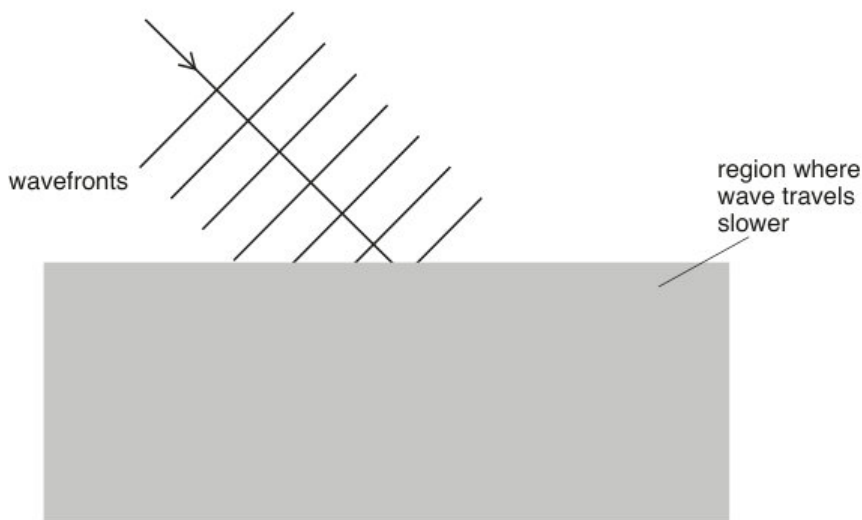


Fig. 10.2

Mark scheme — Q10 (c)(i) only

10(c)(i) ($v = f\lambda$ or $2(000) \times 16$ C1 32 000 cm / s or 320 m / s A1

Question 10 (c)(iii)

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

.....

[3]

Mark scheme — Q10 (c)(iii) only

10(c)(i) ($v = f\lambda$ or $2(000) \times 16$ C1 32 000 cm / s or 320 m / s A1

May/June 5054/22

Question 6 (a)

- 6 (a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

Mark scheme — Q6 (a) only

6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1

Question 6 (b)(i)

- (i) Calculate the wavelength of the sound.

wavelength =[2]

- (ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

[1]

Mark scheme — Q6 (b)(i) only

6(b)(i) ($\lambda = v / f$ or $330 / 3800$ B1 0.087 m or 8.7 cm B1

2016

Oct/Nov 5054/21

Question 5 (a)

- (a) State what is meant by the *frequency* of a sound.

.....

[1]

Mark scheme — Q5 (a) only

- | | | |
|---|---|----------|
| 5 | (a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time | B1 |
| | (b) (i) ($\lambda = v / f$ or $330 / 2200$)
0.15 m | C1
A1 |
| | (ii) 1 no change
and
2 increases | B1 |

5 (a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time B1 (b) (i) ($\lambda = v / f$ or $330 / 2200$) C1 0.15 m A1 (ii) 1 no change and 2 increases B1 (c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) B1 2 compressions and rarefactions / molecules vibrate / longitudinal wave B1 vibration / oscillation / energy passed on B1 (ii) fewer / no molecules /

Question 5 (b)

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

- (ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....[1]

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Mark scheme — Q5 (b) only

Oct/Nov 5054/22

Question 10 (a)

- (a) The red light from the laser has a frequency of 4.3×10^{14} Hz.

- (i) State the speed of light in air.

.....[1]

- (ii) Calculate the wavelength of this red light in air.

wavelength =[2]

Mark scheme — Q10 (a) only

- 10 (a) (i) $3.0 \times 10^8 \text{ m/s}$ B1
- (ii) $(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ C1
 $7.0 \times 10^{-7} \text{ m}$ A1 [3]
- (b) (i) decreases B1
- (ii) $\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 C1
 49° A1
- (iii) 41° R1 [4]

10 (a) (i) $3.0 \times 10^8 \text{ m/s}$ B1 (ii) $(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ C1 $7.0 \times 10^{-7} \text{ m}$ A1 [3] (b) (i) decreases B1 (ii) $\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 C1 49° A1 (iii) 41° B1 [4] (c) (i) dispersion at both surfaces and refractions in correct direction B1 violet / blue light below the red light shown B1 (ii) spectrum or band of (continuous) colours or colours of rainbow B1 red, orange, yellow, green, blue, (indigo, violet) B1 (iii) 1 X marked above red B1 2 it is / black surfaces are good absorbers (of IR radiation) B1 [6] (d) intruder / human being emits IR or IR beam broken or IR reflected B1 intruder warm or IR detected does not reach detector change detected B1 [2] [15]

May/June 5054/21

Question 5 (a)

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

Fig. 5.1

[2]

Mark scheme — Q5 (a) only

- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time/period for one wave(length)/cycle constant B1
or each oscillation/cycle takes one second
- 6 (a) angle of incidence B1

5 (a) *sound*: along or parallel (to transfer of energy or wave) and longitudinal B1 *water*: perpendicular and transverse B1 (b) (i) 0.29 – 0.28 m B1 (ii) time / period for one wave(length) / cycle constant B1 or each oscillation / cycle takes one second

Question 5 (b)

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

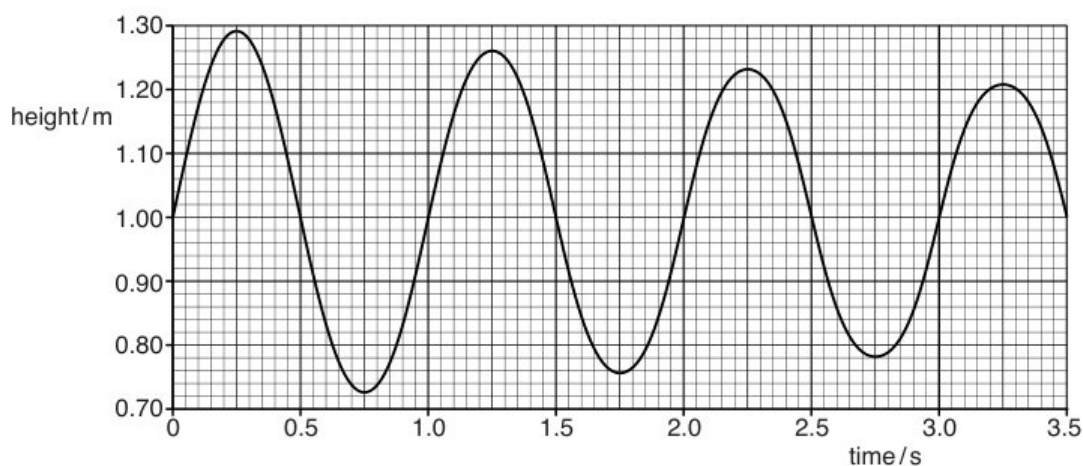


Fig. 5.2

- (i) Determine the amplitude of the vibration at time = 0.25 s.
[1]
- (ii) Describe how the graph shows that the frequency of the wave is constant.

Mark scheme — Q5 (b) only

Question 7 (c)

- (c) The loudspeaker produces sound of frequency 500 Hz. The speed of sound is 320 m/s. Calculate the wavelength of the sound produced.

wavelength =[2]

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Mark scheme — Q7 (c) only

May/June 5054/22

Question 6 (a)

- (a) State which colour in Fig. 6.1 has
- (i) the largest wavelength, [1]
- (ii) the highest frequency. [1]

Mark scheme — Q6 (a) only

6 (a) (i) red	B1
(ii) blue	B1
(b) ANY 2 from (the use must agree with the type)	B4
Microwaves	B1
use – satellite television, telephone, mobile / cell phones;	B1
cooking, heating in a microwave oven, television	
remote, radar, communication	

6 (a) (i) red B1 (ii) blue B1 (b) ANY 2 from (the use must agree with the type) B4 Microwaves B1 use – satellite television, telephone, mobile / cell phones; B1 cooking, heating in a microwave oven, television remote, radar, communication X(-rays) B1 use – hospital use in medical imaging or security imaging, killing cancerous cells, B1 and engineering applications such as detecting cracks in metal, crystallography gamma (rays) B1 use – medical treatment in killing cancerous cells, and engineering applications B1 such as detecting cracks in metal, sterilisation, tracer applications, radiotherapy Page 4 Mark Scheme Syllabus Paper – May/June 2016 5054 22 © Cambridge International Examinations 2016

Question 6 (b)

- (b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component

description of use

.....

.....

2. component

description of use

.....

.....

[4]

Mark scheme — Q6 (b) only

Question 10 (b)

- (b) A microphone is connected to a c.r.o. to display a sound wave.

Fig. 10.2 shows the trace on the c.r.o.

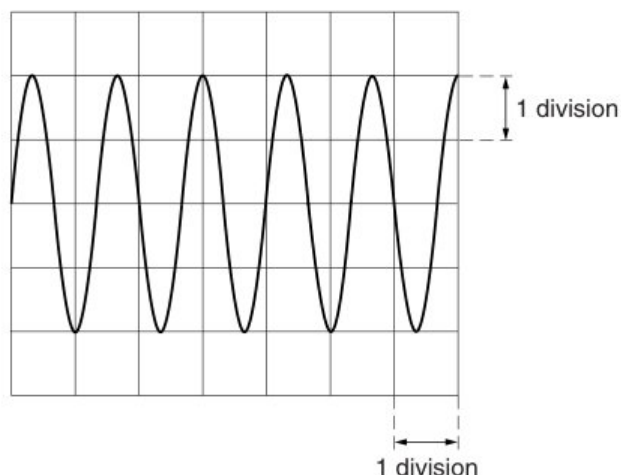


Fig. 10.2

The settings on the c.r.o. are: Y-gain 0.5V/division; time base 2.0ms/division.

- (i) Determine
1. the maximum voltage,

Mark scheme — Q10 (b) only

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Question 7 (a)

- (a) Describe one difference between a compression and a rarefaction.

.....

 [1]

Mark scheme — Q7 (a) only

- | | | | |
|---|-----|--|---|
| 7 | (a) | at compression: molecules closer together or pressure higher or vice versa for rarefaction | B1 |
| | (b) | (i) $v = f\lambda$ or in words | B1 |
| | | (ii) larger and because the frequency is lower | B1 |
| | (c) | states one use
basic idea | (e.g. prenatal scanning) B1
(e.g. ultrasound reflects off foetus) B1 [5] |

7 (a) at compression: molecules closer together or pressure higher or vice versa for rarefaction B1 (b) (i) $v = f\lambda$ or in words B1 (ii) larger and because the frequency is lower B1 (c) states one use (e.g. prenatal scanning) B1 basic idea (e.g. ultrasound reflects off foetus) B1 [5]

Question 7 (b)

(b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

(i) State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
 [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....

 [1]

Mark scheme — Q7 (b) only

7 B1 (ii) 35 B1 [2] (c) (i) two separate sources: rocks (e.g. radon), outer space (e.g. cosmic rays), man-made sources (e.g. nuclear waste / fallout) B2 (ii) 22 counts / minute B1 (iii) 27 counts / minute B1 (iv) use of $27 / 2$ or $27 / 4$ or $27 / 8$ B1 from 85 to 90 days (inclusive) B1 [6] (d) (i) (background count-rate is) reduced B1 not to zero or not stopped or (some) gamma-rays in background count B1 (ii) not sensible M1 all the beta-radiation would be absorbed or no beta-radiation reaches the detector A1 [4] [15]

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Question 4 (a)

(a) Describe the movement of the particles of the string.

.....

[2]

Mark scheme — Q4 (a) only

4	(a)	oscillate / vibrate stated or described	B1
		transverse movement described	B1
	(b)	0.40 m	B1
	(c) (i)	$v = f \lambda$ or $(f =) v / \lambda$ or 2 / (b)	C1
		5.0 Hz	A1
	(ii)	clear attempt to draw wave moved along 0.20 m to right	B1

4 (a) oscillate / vibrate stated or described B1 transverse movement described B1 (b) 0.40 m B1 (c) (i) $v = f \lambda$ or $(f =) v / \lambda$ or 2 / (b) C1 5.0 Hz A1 (ii) clear attempt to draw wave moved along 0.20 m to right B1

Question 4 (b)

- (b) Determine the wavelength of the wave.

wavelength =[1]

Mark scheme — Q4 (b) only

4 B1 α has proton number 2 B1 correct proton number for U ecf their value for α B1 (c) (i) alpha particles only travel a short distance in air B1 or alpha particles stopped / scattered / deflected by air or alpha particles ionise air (ii) particles come off in different directions B1 or not emitted in one line / as a ray or not all the particles pass through the slit (iii) B correct shape and deflected more than A B1 (iv) particles close to / fired at the nucleus are deflected (back) / repelled B1 some particles pass (straight) through B1 a few particles come back / large deflection or most pass (straight) through (with little deviation) B1 and how this explains the nucleus is small

Question 4 (c)

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

frequency =[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

Mark scheme — Q4 (c) only

May/June 5054/22

Question 9 (a)

- (a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

Mark scheme — Q9 (a) only

9 (a)	number of waves (that pass a point) or number of oscillations (passing a point) in unit time or per second or in 1 second	M1 A1
(b) (i)	1.5 cm	B1
(ii)	$(v =)f\lambda$ or 5×1.5 seen or $(s=)d/t$ and $f = 1/t$ 7.5 cm/s	C1 A1

9 (a) number of waves (that pass a point) M1 or number of oscillations (passing a point) in unit time or per second or in 1 second A1 (b) (i) 1.5 cm B1 (ii) $(v =)f\lambda$ or 5×1.5 seen C1 or $(s=)d/t$ and $f = 1/t$ 7.5 cm / s A1 (c) (i) wavelength decreases B1 travels a shorter distance in the same time B1 or frequency stays the same (and $v = f\lambda$) Page 5 Mark Scheme Syllabus Paper – May/June 2015 5054 22 © Cambridge International Examinations 2015 (ii) wavefronts with smaller wavelength (by eye) B1 smaller angle to surface (by eye) and slanted down B1 wavefronts join those in shallow water B1 (d) (i) sound water B1 B1 one row only particles / wave / source • vibrate / oscillate / move in direction of (travel of) wave / along wave • move backwards and forwards particles / wave / source • vibrate / oscillate / move at 90° to direction of (trav

Question 9 (b)

- (b) (i) Determine the wavelength of the wave in deep water.

..... [1]

- (ii) Calculate the speed of the wave in deep water.

speed = [2]

Mark scheme — Q9 (b) only

Question 9 (c)

- (c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

- (i) State and explain how this affects the wavelength of the wave.

.....

 [2]

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Mark scheme — Q9 (c) only

Question 9 (d)

(d) Sound is also a wave.

(i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

(ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

Mark scheme — Q9 (d) only
